



THE UNIVERSITY OF
MELBOURNE

SQL Part 3

Database Systems & Information Modelling
INFO90002.

Week 5 – SQL

Presented by Greg Wadley

Content by: Dr Tanya Linden, Dr Renata Borovica-Gajic, David Eccles, Dr Greg Wadley





This Lecture Objectives

Extending your knowledge

Query Nesting, Comparison Operators, EXISTS, NOT EXISTS, VIEWS

- DML
 - Query nesting
 - Subquery
 - Views
- DCL
 - Data Control Language

How to think about SQL

- Problem Solving



Query Nesting

Complex queries



Query Nesting

SQL provides the ability to *nest* subqueries

A nested query is simply another select query you write to produce a table set

- Remember that all **SELECT** queries return a table set of data

A common use of subqueries is to perform set tests

- Set membership, set comparisons



Sub-Query Comparison Operators

IN / NOT IN

- Used to test whether the attribute is IN/NOT IN the subquery list

ANY

- True if any value returned meets the condition

ALL

- True if all values returned meet the condition

EXISTS

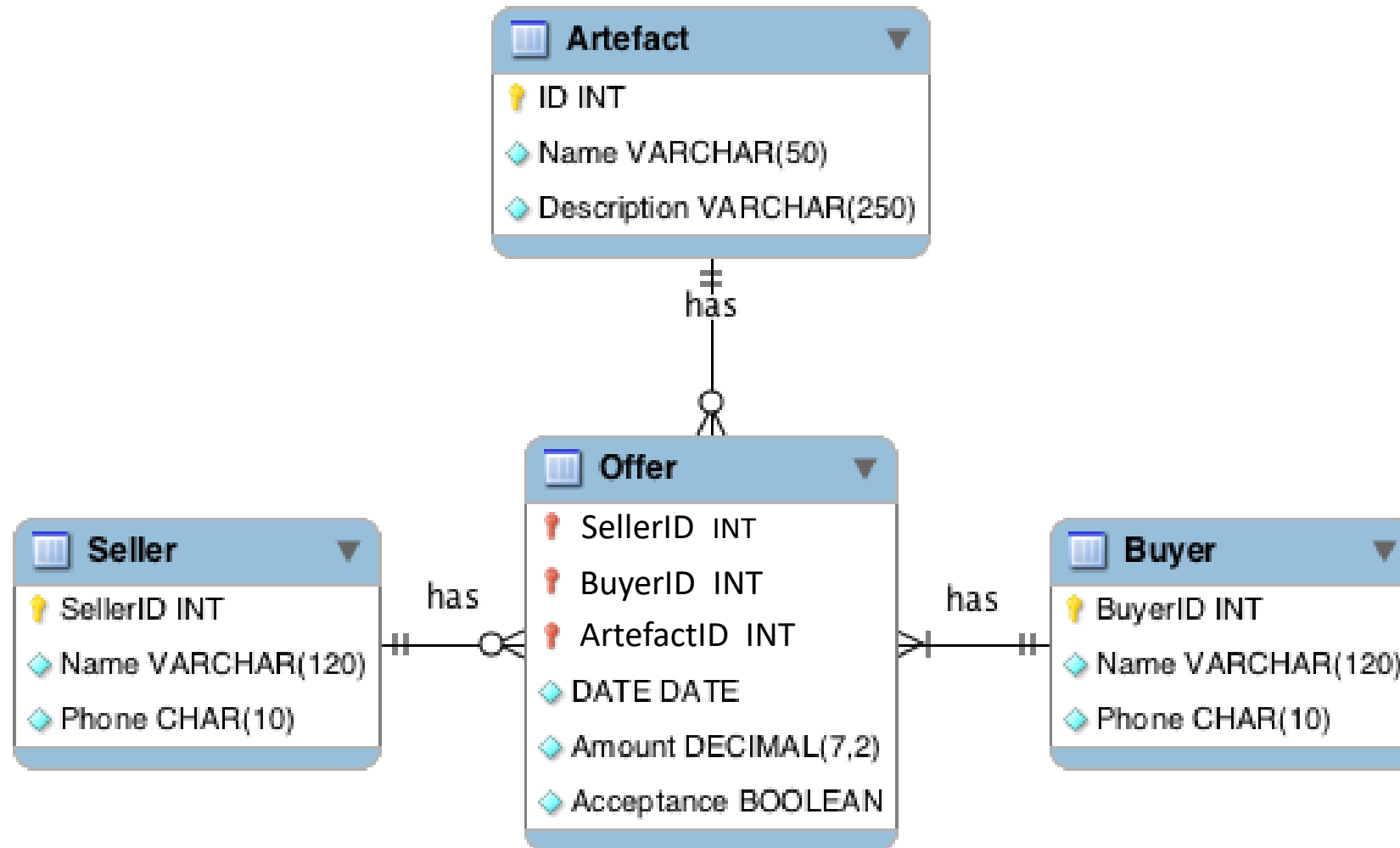
- True if the subquery returns one or more records

For more info:

- https://www.w3schools.com/sql/sql_any_all.asp
- https://www.w3schools.com/sql/sql_exists.asp

General help with SQL: <https://www.w3schools.com/sql/> (great tutorial)

Auction Bids – Physical Model



Tables

Seller

	SellerID	Name	Phone
▶	1	Ann	0509 123 321
	2	Bill	0518 234 432
	3	Carol	02 8344 4777

Buyer

	BuyerID	Name	Phone
▶	1	Maggie	0539 335 577
	2	Nigel	0519 434 389
	3	Olga	13 24 35

Artefact

	ID	Name	Description
▶	1	Vase	Ming Vase H50cm W30cm
	2	Sketch	Early Modern Dutch School
	3	Pot	CopperUS 18th Century

Offer

	SellerID	BuyerID	ArtefactID	OfferDate	Ammount	Acceptance
▶	1	1	1	2012-06-20	81223.23	0
	1	1	2	2012-06-20	82223.23	0
	2	2	1	2012-06-20	1995.50	0
	2	2	2	2012-06-20	2300.15	0

Example: Subquery

List the BuyerID, Name and Phone number for all bidders on artefact 1

```
1 • SELECT *
2 FROM Buyer
3 WHERE buyerID IN
4   (SELECT BuyerID
5    FROM Offer
6    WHERE artefactid = 1);
```

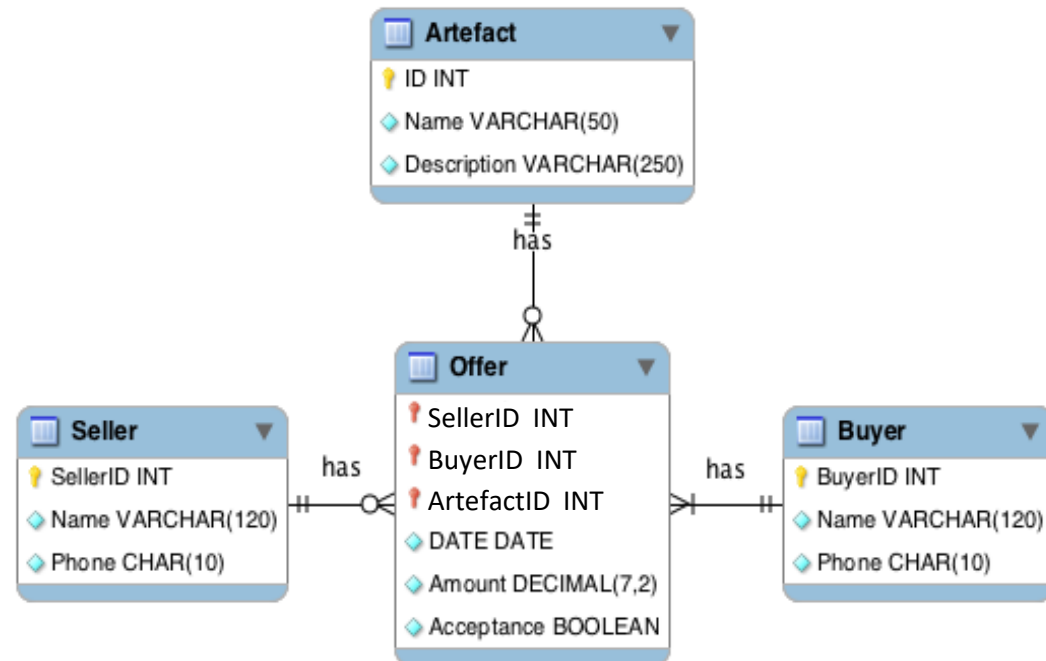
	BuyerID	Name	Phone
►	1	Maggie	0539 335 577
	2	Nigel	0519 434 389

More examples using subqueries

Which Artefacts don't have any offers made on them

```
SELECT *
FROM artefact
WHERE ID NOT IN
    (SELECT artefactid
     FROM offer);
```

ID	Name	Description
3	Pot	CopperUS 18th Century





Exercise

Which Buyers haven't made a bid for the "Pot" Artefact ?

Exercise – Solution

- Which buyer haven't made a bid for the "Pot" Artefact ?

```
SELECT *  
FROM buyer  
WHERE buyerID NOT IN  
  (SELECT BuyerID  
   FROM offer  
   WHERE artefactID IN  
     (SELECT ID  
      FROM artefact  
      WHERE name = 'Pot'));
```

1	Maggie	0539 335 577
2	Nigel	0519 434 389
3	Olga	13 24 35



This or That? Writing better queries

- Do we need to use IN? Is there another way...
- List all buyer details for all bidders on artefact 1

```
SELECT *  
FROM Buyer  
WHERE BuyerID IN  
  (SELECT BuyerID  
   FROM Offer  
   WHERE ArtefactID = 1)
```

Is equal to:

```
SELECT *  
FROM Buyer b  
INNER JOIN Offer o  
ON b.BuyerID = o.BuyerID  
WHERE ArtefactID = 1
```



This is a more efficient way

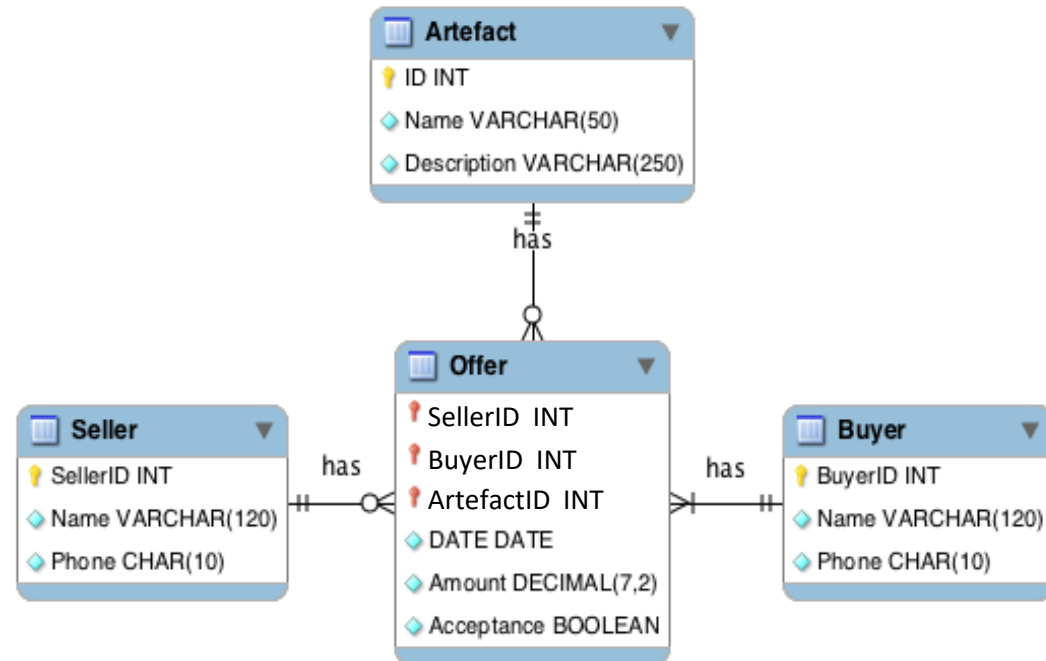
EXISTS - example

Returns true if the subquery returns one or more records

Example: *List all buyer details for all bidders on artefact 1*

```
SELECT * FROM Buyer
WHERE EXISTS
  (SELECT * FROM Offer
   WHERE Buyer.BuyerID = Offer.BuyerID
   AND ArtefactID = 1)
```

BuyerID	Name	Phone
1	Maggie	0333333333
2	Nicole	0444444444





Views

SQL DDL and DML



Views

Any relation that is not in the physical models, but is made available to the “user” as a virtual relation is called a view.

Views are good because:

- They help hide the query complexity from users
- They help hide data from users (e.g. data users are not authorised to see)
 - Different users use different views
 - Prevents customers from seeing quantities in stock or original purchase price of an item, for instance
 - One way of improving database security

Syntax:

CREATE VIEW nameofview **AS** validsqlstatement

Once a view is defined,

- its definition is stored in the database (not the data, but metadata – schema information)
- it can be used just like any other table
- Every time shows up-to-date data

Create View Example

```
CREATE VIEW EmpPay AS
SELECT Employee.ID, Employee.Name, DateHired,
       EmployeeType, HourlyRate AS Pay
FROM Employee INNER JOIN Hourly
ON Employee.ID = Hourly.ID

UNION
SELECT Employee.ID, Employee.Name, DateHired,
       EmployeeType, AnnualSalary AS Pay
FROM Employee INNER JOIN Salaried
ON Employee.ID = Salaried.ID

UNION
SELECT Employee.ID, Employee.Name, DateHired,
       EmployeeType, BillingRate AS Pay
FROM Employee INNER JOIN Consultant
ON Employee.ID = Consultant.ID;
```


Using a View

SELECT * FROM EmpPay;

ID	Name	DateHired	EmployeeType	Pay
3	Alice	2012-12-02	H	23.43
4	Alan	2010-01-22	H	29.43
1	Sean	2012-02-02	S	92000.00
2	Linda	2011-06-12	S	92300.00
5	Peter	2010-09-07	C	210.00
6	Rich	2012-05-19	C	420.00

**SELECT * FROM EmpPay
WHERE EmployeeType = "H" OR EmployeeType = "C"**

ID	Name	DateHired	EmployeeType	Pay
3	Alice	2012-12-02	H	23.43
4	Alan	2010-01-22	H	29.43
5	Peter	2010-09-07	C	210.00
6	Rich	2012-05-19	C	420.00

Output Snippets Query 1 Result Lecture7.sql Result x
 Fetched 4 records. Duration: 0.000 sec, fetched

CREATE VIEW example

```
CREATE VIEW v_DEPT_SALARY AS
SELECT department.departmentid, department.name, sum(employee.salary)
AS DEPTSAL
FROM department
INNER JOIN employee
ON department.departmentid = employee.departmentid
GROUP BY department.departmentid, department.name;
```



Data Control Language

SQL DCL



Data Control Language / Other Commands

DCL

- Users and permissions
CREATE USER, DROP USER
GRANT, REVOKE
SET PASSWORD

Other Commands

- Database administration
BACKUP TABLE, RESTORE TABLE
ANALYZE TABLE
- Miscellaneous
DESCRIBE tablename
USE db_name

They are typically called 'Database Administration Statements'



How to think like SQL...

SQL DDL and DML



How to Think like SQL

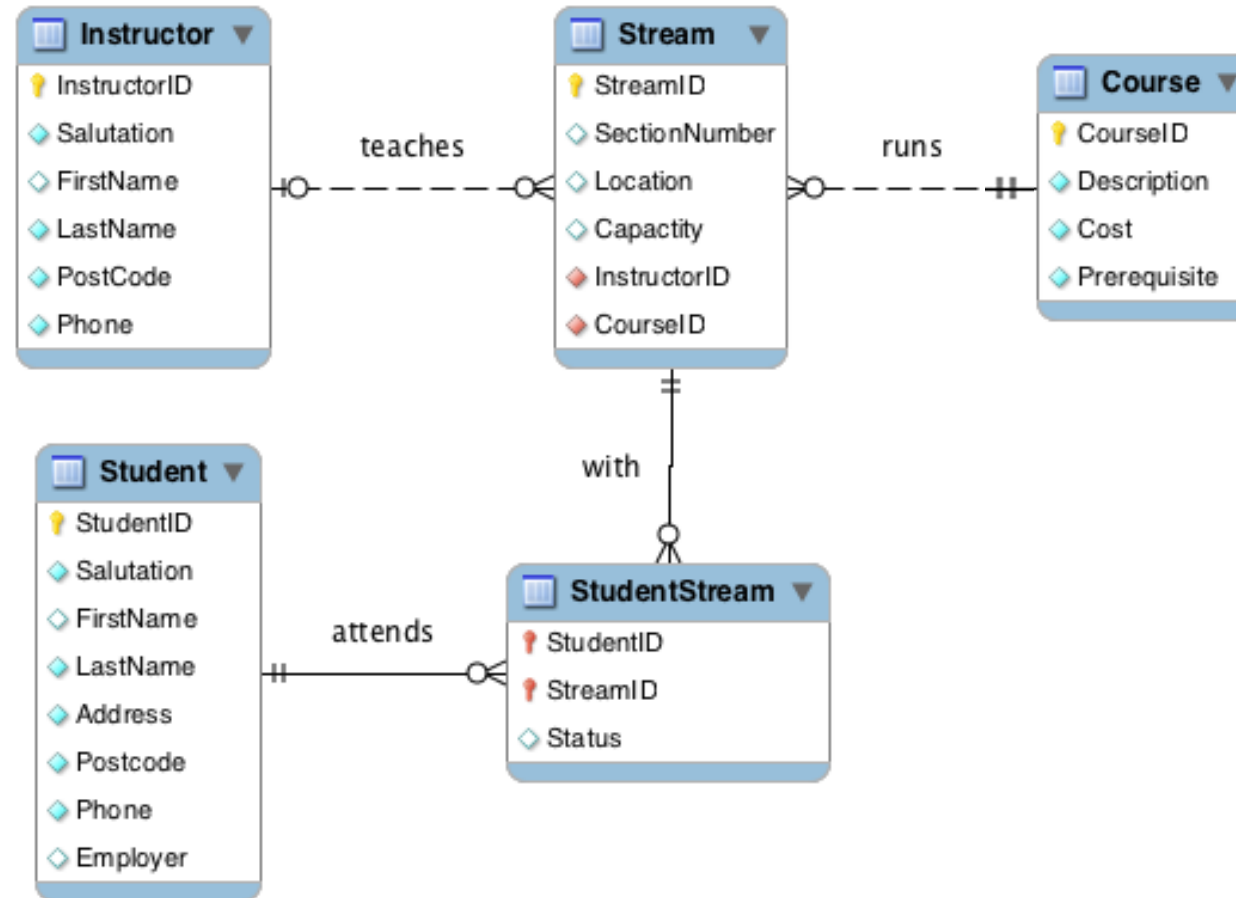
It's going to be critical for you to think like SQL to handle the queries you will need to write...

Hopefully the following discussion will help you in this endeavour:

1. **USE** the database design as a **MAP** to help you when you are formulating queries
2. **USE** the structure of the **SELECT** statement as a template
3. **FILL** out parts of the **SELECT** structure and **BUILD** the query

Let's try it!

Example



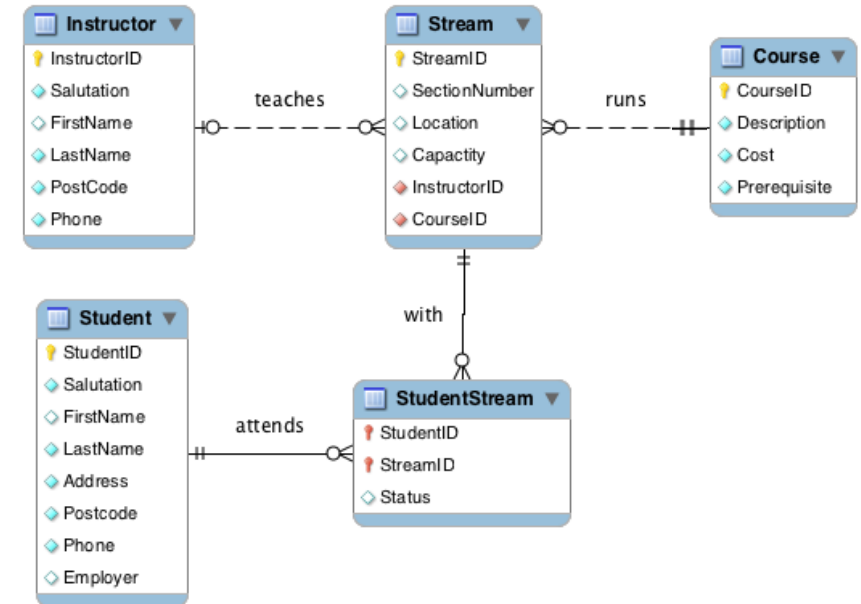
Example: Which employers employ students who are doing a course in locations where the capacity is greater than 20 persons, and what are those locations?

How to approach writing queries

Which employers employ students who are doing a course in locations where the capacity is greater than 20 persons, and what are those locations?

What is the query asking for:

- Which fields and tables:
Fields: Employer, Location, Capacity
Tables: Student, Stream, StudentStream
- But only if the capacity > 20 (condition)



Let's try to use the structure of the SELECT statement now:

```

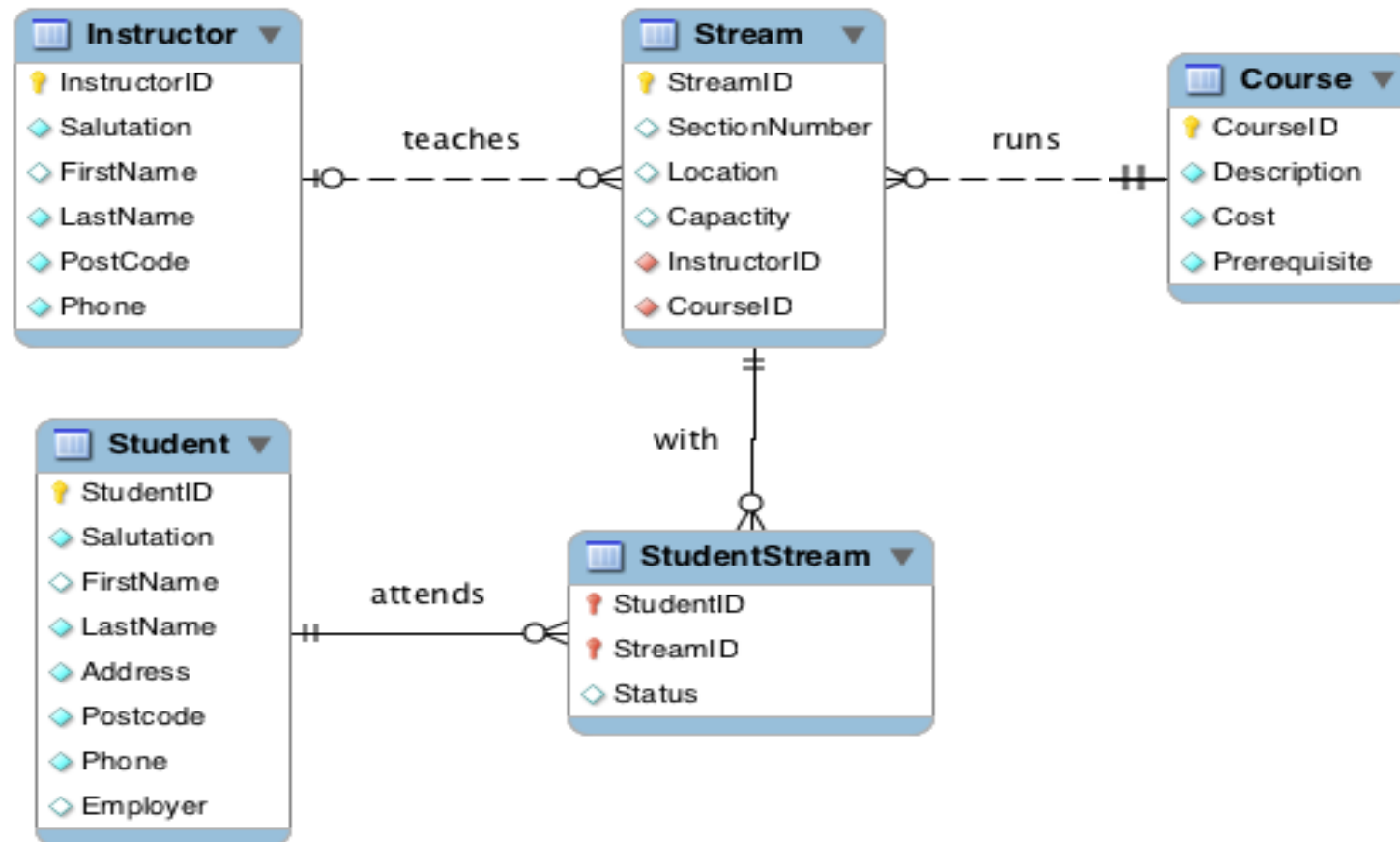
SELECT Employer, Location, Capacity
FROM Student INNER JOIN StudentStream
INNER JOIN Stream
ON Student.StudentID = StudentStream.StudentID
AND Stream.StreamID = StudentStream.StreamID
WHERE Capacity > 20;
    
```

```

SELECT Employer, Location, Capacity
FROM Student
INNER JOIN StudentStream
ON Student.StudentID = StudentStream.StudentID
INNER JOIN Stream
ON Stream.StreamID = StudentStream.StreamID
WHERE Capacity > 20;
    
```


Exercise (your turn)

What is the phone number of the instructor who teaches a course that costs over \$10000 attended by studentID 202.





SQL Video

A very good overview of SQL:

<https://www.youtube.com/watch?v=uRdIdd-UkTc&index=7&list=PLdQddgMBv5zHcEN9RrhADq3CBColhY2hl>

(It runs for 90+ minutes so for viewing in your own time)

Additional practice exercises on LeetCode

Easy to Medium <https://leetcode.com/studyplan/top-sql-50/>



What's examinable

- **You need to know how to write SQL**
 - **DML**
 - **DDL**



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